

MLOps

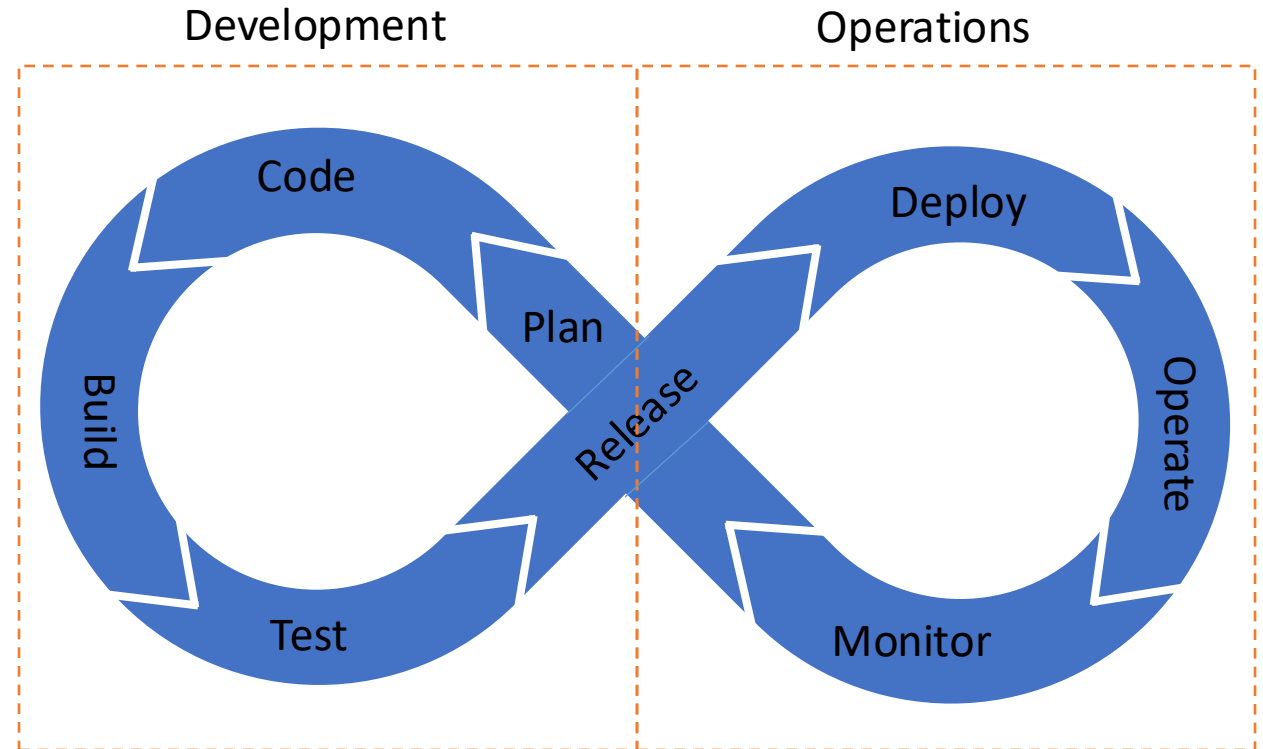
What Is MLOps? - MLOps Persona - Create a Plan

Session 1

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DevOps

- A mindset for collaborating between developers and operations
- A software development method that stresses communication, collaboration and integration between SW developers and IT professionals:
 - Enable rapid evolution of products and services
 - Reduce risk, improve quality across portfolio and reduce costs
- A set of practices intended to **reduce the time between committing a change to a system and the change being placed into normal production, while ensuring high quality**



What is CI/CD?

- Method to frequently put apps into production by introducing automation into the stages of app development
- Main concepts:
 - Continuous Integration
 - Continuous Delivery (Continuous Deployment)



Continuous Integration (CI) focuses on integrating code changes by developers so that the main branch is up-to-date



Needs a Version Control System like Git Repository

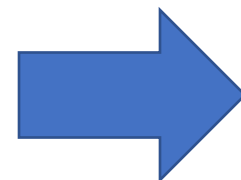
Continuous Delivery (CD) focuses on taking the results of continuous integration builds and put them into 'production like' environment (Staging Environment) and do some tests, then deploy to final production environment



Make sure that the build is in a 'deployable state'

Why Do We Need MLOps?

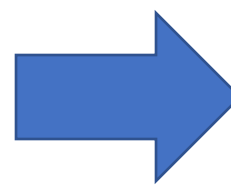
- Managing ML life cycles at scale is challenging:
 - There are many dependencies
 - Data constantly changing
 - Business needs shift
 - Not everyone speaks the same language
 - Business
 - Data
 - Modeling
 - IT
 - Data scientists are not software engineers
 - Mostly focused on model development
 - Complexity becomes exponential
 - Many models in production
 - Turnover of staff



This is similar to DevOps scenarios

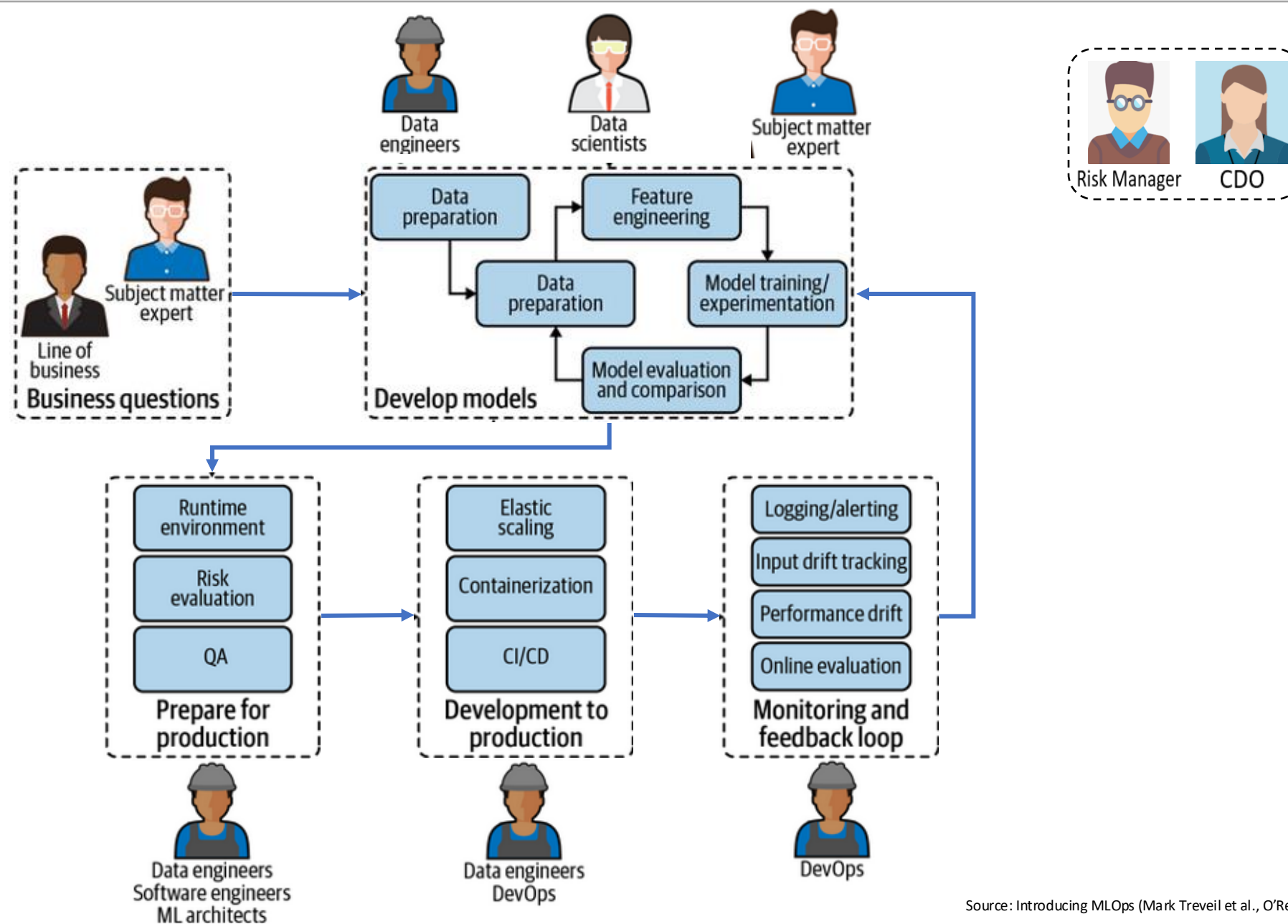
What Is The Difference?

- ML Models are not static!!!!!!!!!!!!
- Examples:
 - Credit Model in Banking (Bias, Customer behavior, Changes in customer base)
 - Product Recommendation in TV Platform (Customer behavior, Changes in offering)
 - Sales Forecast in Retail (Changes in customer behavior, Economic situation)



ML Models need retraining

Persona



Source: Introducing MLOps (Mark Treveil et al., O'Reilly, 2020)

Relationships

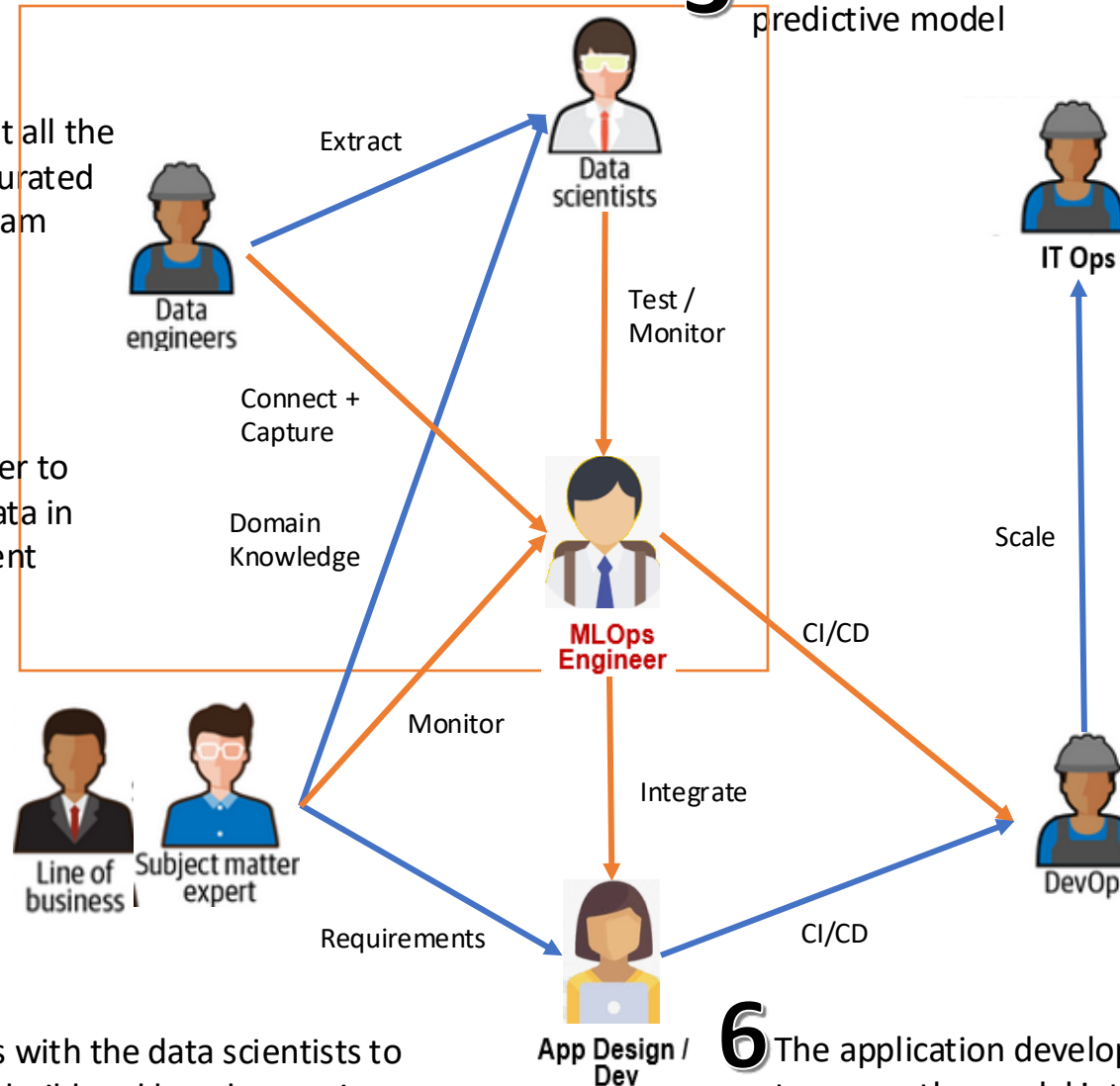
2 Works on the databases to ensure that all the sources are available and to provide curated datasets to the rest of the analytics team

4 Then they work with the MLOps engineer to test the trained model on production data in a development or “sandbox” environment

1 Business users and Subject matter expert give domain knowledge and requirements

5 The application designer works with the data scientists to decide which models to use or build and how best to integrate the model so that it is useful to business users

3 Explore the data, train, and build the predictive model



9 IT operations, which maintains the infrastructure and ensures that the application and its models run, perform, and scale optimally.

7 The MLOps engineer takes that model and deploys it into production environments (e.g., first into pre-production, then into final production)

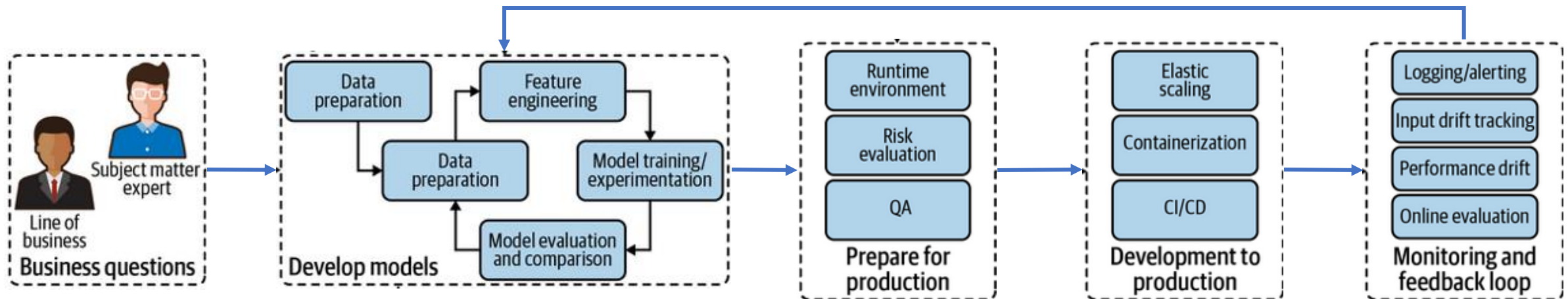
8 DevOps team smooths the transition from development to IT operations

6 The application developer writes the code to weave the model into the application



Another pain

- Many ML projects are doomed from the start due to a misalignment between product metrics and model metrics.
- Other projects fail by producing good models that simply aren't helpful for the business
- For each successful result published in a research paper or corporate blog, there are hundreds of reasonable-sounding ideas that have entirely failed
- Is it possible to create a plan?



4 categories of performance

Business
Performance

Model
Performance

Data
Performance

System
Performance

Business Performance

- Once the goal is clear, a **metric** should be defined to judge it **success**
- Example:

Goal: Online Product Recommendations on a Web Site

Product metrics

- Increase Click-through Rate
- Number of users that a product attracts

Guardrail metrics (metrics that shouldn't decline below a given point)

- Average user session length



Remember: product metrics that impact business performance are **the only ones that matter**

Model Performance

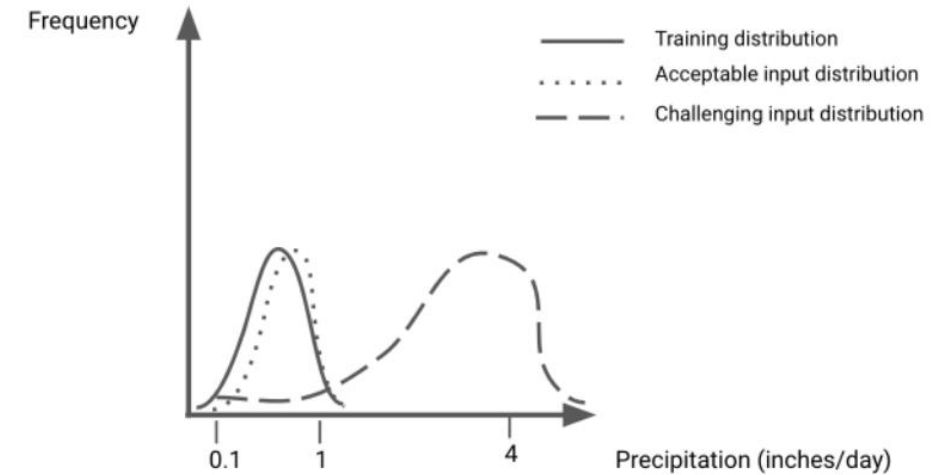
- Different modeling approaches use different model metrics:
 - In classification: accuracy, area under ROC curve, F1 score, ...
 - In regression: root mean squared error, mean absolute error, R2, ...
- But in our example:
 - Model performance: How many people clicked on recommended product
- Finally, a product/service/app should be designed with reasonable assumptions of model performance in mind.
- If a product relies on a model being perfect to be useful, it is very likely to produce inaccurate or even **dangerous** results.



Remember: we need the level of performance that is **sufficient** to accomplish our goal

Data Performance

- Models rely on data
- Most models need to have been exposed to training data that is similar to a given input to perform well
- What happens if there are significant changes in the data distribution?
- Not all problems have the same requirements



Depending on your business problem, you should consider how hard it will be to keep models fresh

How often will you need to retrain models, and how much will it cost you each time we do so?

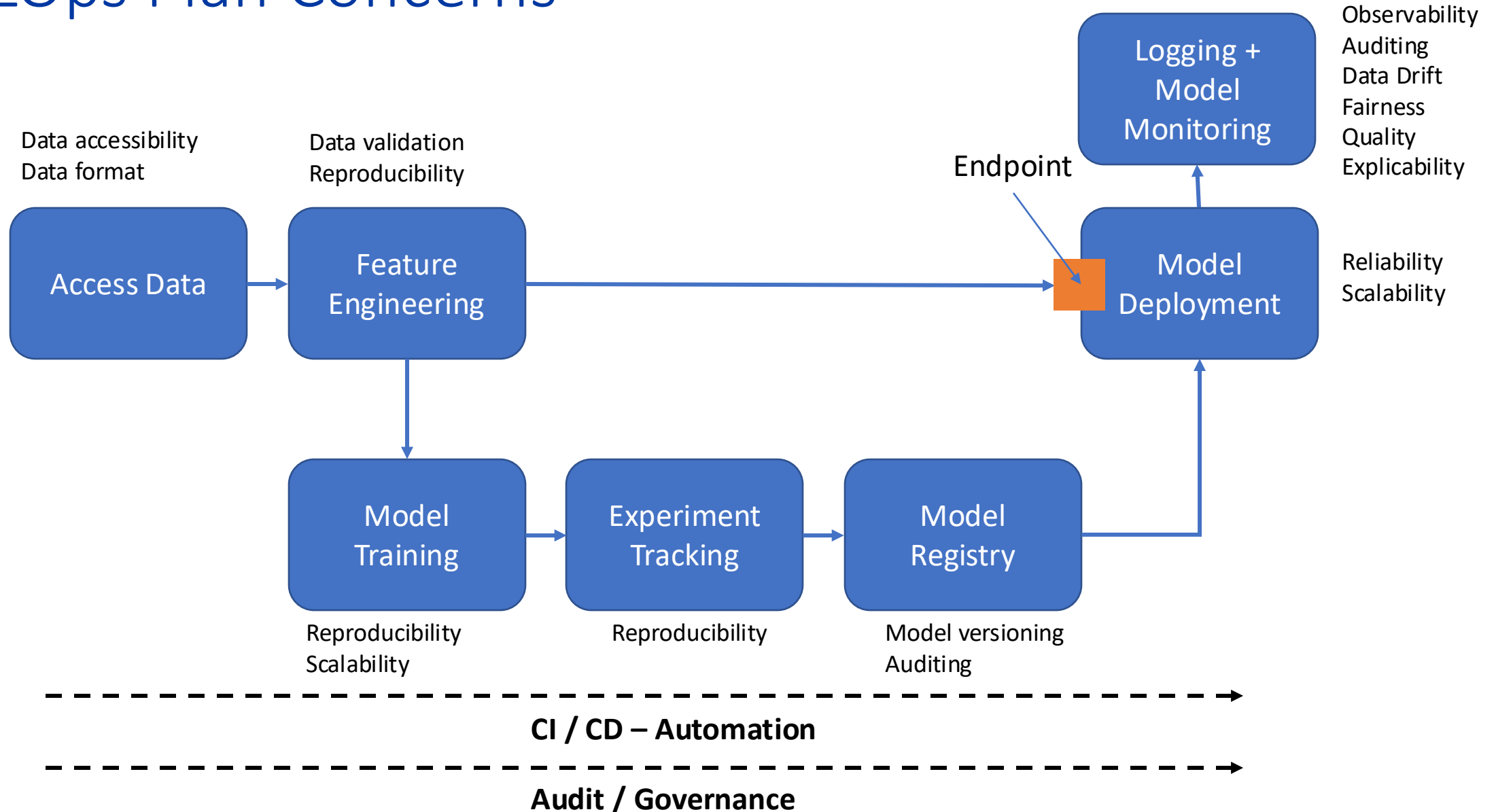
System Performance

- Ideally, a model should deliver a prediction quickly. But how quickly is quickly?
- Model inference runtimes increase as models get more complex (**and costs!!!**)
- Depending on your problem, there are other categories you could consider, such as hardware constraints, development time, and maintainability



Remember: don't confuse the performance of a model in development with the performance of a model in production

MLOps Plan Concerns



CI/CD with Gitlab Git Repository